PREDICTING INDUSTRIAL TRUCK FAILURES FROM OPERATIONAL TELEMETRY

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BY

B SATHWIK	**24071A3204**
B SAI CHARAN	**24071A3206**
K JAYANTH VARMA	**24071A3232**
MD SAMEER	**24071A3243**
P AKSHAY KUMAR	**24071A3247**

UNDER THE GUIDANCE OF

Mr. K. Akhil
Assistant Professor, Department of CSE

Mrs. G. Lakshmi Deepthi
Assistant Professor, Department of CSE



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
VALLURUPALLI NAGESWARA RAO
VIGNANA JYOTHI INSTITUTE OF ENGINEERING & TECHNOLOGY
Vignana Jyothi Nagar, Bachupally, Hyderabad 500118, Telangana, India

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VALLURUPALLI NAGESWARA RAO VIGNANA JYOTHI INSTITUTE OF ENGINEERING & TECHNOLOGY
AUTONOMOUS INSTITUTION

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CERTIFICATE

This is to certify that the CBP Project titled "**PREDICTING INDUSTRIAL TRUCK FAILURES FROM OPERATIONAL TELEMETRY**" is a record of bonafide work submitted by:

B SATHWIK	**24071A3204**
B SAI CHARAN	**24071A3206**
K JAYANTH VARMA	**24071A3232**
MD SAMEER	**24071A3243**
P AKSHAY KUMAR	**24071A3247**

in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology (B.Tech.) in **COMPUTER SCIENCE AND BUSINESS SYSTEMS** of **Jawaharlal Nehru Technological University Hyderabad (JNTUH)**, under **VNRVJIET B.Tech. R25 Academic Regulations**. The work has been carried out under our guidance and supervision at Vallurupalli Nageswara Rao Vignana Jyothi Institute of Engineering & Technology, Hyderabad.

The results embodied in this report have not been submitted to any other University or Institute for the award of any degree or diploma. The report is original, free from plagiarism, and complies with Institute guidelines on academic ethics.

Signature of Supervisor	**Signature of Supervisor**	**Signature of HOD**
Mr. K. Akhil	Mrs. G. Lakshmi Deepthi	Dr. V Baby
Assistant Professor	Assistant Professor	HOD
Department of CSE	Department of CSE	Department of CSE
VNR VJIET, Hyderabad	VNR VJIET, Hyderabad	VNR VJIET, Hyderabad

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AUTONOMOUS INSTITUTION

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

DECLARATION

We, the undersigned, declare that the Project titled "**PREDICTING INDUSTRIAL TRUCK FAILURES FROM OPERATIONAL TELEMETRY**" has been carried out by us in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology (B.Tech.) in **COMPUTER SCIENCE AND BUSINESS SYSTEMS** of **Jawaharlal Nehru Technological University Hyderabad (JNTUH)**, under **VNRVJIET B.Tech. R25 Academic Regulations**, at Vallurupalli Nageswara Rao Vignana Jyothi Institute of Engineering & Technology, Hyderabad.

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PLACE: HYDERABAD

DATE: _____

SIGNATURE OF THE STUDENT(S)

B SATHWIK	**24071A3204**
B SAI CHARAN	**24071A3206**
K JAYANTH VARMA	**24071A3232**
MD SAMEER	**24071A3243**
P AKSHAY KUMAR	**24071A3247**

ABSTRACT

Heavy-duty industrial vehicles form the critical foundation of modern global logistics, mining, haulage, and supply-chain infrastructure. Unexpected mechanical breakdowns—particularly catastrophic failures in mission-critical powertrain sub-assemblies such as Component X—induce severe secondary physical damage, unscheduled logistics bottlenecks, and exorbitant emergency repair costs. Conventional maintenance paradigms remain sharply bifurcated into reactive post-breakdown repairs and conservative schedule-based servicing. While preventive maintenance reduces catastrophic road failures, it frequently discards mechanical components possessing substantial remaining useful life, generating substantial operational expenditure. Condition-based predictive maintenance, by contrast, leverages high-frequency operational telemetry streams to forecast impending component degradation well before functional breakdown occurs.

This Community-Based Project (CBP) presents the design, mathematical formulation, leak-free implementation, and comprehensive empirical evaluation of a condition-based predictive maintenance architecture for heavy-duty industrial vehicles based on the standardized SCANIA Component X benchmark schema. The operational telemetry encompasses cumulative wear counters, multidimensional torque-speed histograms, and categorical vehicle chassis specifications. To guarantee zero temporal data leakage, we introduce a strictly causal preprocessing pipeline incorporating chronological sorting per vehicle, forward-fill imputation, first-order differencing ($\Delta t = x_t - x_{t-1}$), non-linear signed-log transformations ($g(\Delta) = \text{sign}(\Delta) \cdot \ln(1 + |\Delta|)$) to tame heavy-tailed increments, and a RobustScaler fitted strictly on the training partition.

From this causal representation, we engineer 210 temporal channels capturing instantaneous readouts, vehicle operational age, and causal rolling statistics (mean, standard deviation, and maximum) across multiple window horizons ($W \in \{6, 12, 24\}$ timesteps). A LightGBM gradient boosted decision tree classifier is trained to forecast failure events within a forward horizon of $H = 48$ timesteps across 400 mutually exclusive commercial vehicles (250 training, 75 validation, and 75 held-out test trucks). Recognizing the profound asymmetry of industrial maintenance economics, decision thresholds are optimized against the official Scania Cost Matrix ($500 per missed failure versus $10 per proactive inspection). On the held-out test split (8,083 observations), the optimized model achieves an exceptional PR-AUC of 0.7893, ROC-AUC of 0.9303, Recall of 77.60%, Precision of 50.85%, and F1-score of 0.6144 at an optimal threshold of $\theta^* = 0.010$. Crucially, cost-sensitive thresholding reduces total fleet maintenance expenditure from $64,860 (standard 0.50 threshold) to $24,430, delivering a dramatic 62.3% financial saving ($40,430 operational reduction). The architecture is integrated into a production-grade FastAPI server and web dashboard.

Keywords: Predictive Maintenance, Industrial Telemetry, Causal Feature Engineering, LightGBM, Scania Cost Matrix, Asymmetric Loss.

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CHAPTER 1 – INTRODUCTION

1.1 Background and Motivation

Modern heavy-duty commercial transport and industrial haulage vehicles operate under severe mechanical, thermal, and operational stress regimes. In long-haul logistics, surface mining, and distributed freight fleets, vehicle reliability directly governs operating profitability, driver safety, and contract fulfillment. Among the numerous sub-assemblies comprising a heavy commercial truck, the powertrain and its mission-critical sub-components (designated generically as Component X within the standardized SCANIA industrial challenge framework) represent the primary sources of catastrophic in-service mechanical breakdowns. Catastrophic powertrain failure during transport incurs extreme direct financial costs, including highway towing charges, emergency field repairs, secondary mechanical damage, driver downtime, and severe penalties for delayed delivery.

Historically, industrial fleet maintenance has been dominated by two contrasting paradigms: reactive post-breakdown repairs and conservative time- or mileage-based preventive servicing. Reactive maintenance operates on a run-to-failure model, resulting in maximum component utilization at the expense of unpredictable breakdowns and catastrophic collateral damage. Preventive maintenance attempts to avoid breakdown by scheduling inspections and component overhauls at predetermined operational intervals (e.g., every 50,000 kilometers). While preventive maintenance reduces road failures, it discards substantial remaining useful life (RUL) in components operating under favorable environmental conditions. Condition-based predictive maintenance (PdM) resolves this fundamental dilemma by continuously monitoring vehicle telemetry to detect subtle precursor degradation patterns, enabling proactive intervention precisely when failure becomes imminent.

1.2 Problem Statement & Industrial Context

Predicting industrial truck failures from operational telemetry presents formidable statistical and engineering challenges. Modern commercial vehicles stream heterogeneous sensor signals, including cumulative operational counters (e.g., total distance traveled, accumulated engine hours, lifetime brake applications) and multidimensional operational histograms (binned distributions of engine speed, torque demand, coolant temperature, and exhaust pressure). These telemetry streams exhibit severe real-world anomalies: irregularly sampled readouts, right-censored vehicle operational histories, extreme class imbalance where component failure events represent less than 3% of all telemetry observations, and pervasive risks of temporal data leakage.

Furthermore, industrial maintenance economics are profoundly asymmetric. A false positive prediction (recommending a proactive workshop inspection for a healthy truck) incurs minor labor overhead ($10), whereas a false negative prediction (missing an impending catastrophic breakdown) triggers road failure, cargo loss, and emergency overhauls ($500). Standard machine learning algorithms optimizing symmetric loss metrics such as accuracy or standard cross-entropy exhibit severe failure in this domain. This project addresses the challenge of building a mathematically sound, leak-free, cost-sensitive predictive maintenance pipeline that minimizes the total operational cost function.

1.3 Research Objectives

The overarching objective of this Community-Based Project is to construct, validate, and deploy an end-to-end machine learning system for predicting commercial vehicle Component X failures. Specific technical objectives include:

- **Rigorous Causal Preprocessing:** Establish a leak-free temporal feature engineering pipeline utilizing chronological vehicle sorting, causal forward-fill imputation, first-order differencing, and signed-log compression to handle heavy-tailed sensor increments.
- **Multi-Window Temporal Feature Representation:** Extract 210 degradation features capturing instantaneous readouts, vehicle operational age, and causal rolling statistics (mean, standard deviation, and maximum) across windows $W \in \{6, 12, 24\}$ timesteps.
- **Strict Vehicle-Level Partitioning:** Partition a fleet of 400 commercial vehicles into mutually exclusive training (250 trucks), validation (75 trucks), and held-out test (75 trucks) splits, guaranteeing zero data leakage across split boundaries.
- **Cost-Sensitive Asymmetric Optimization:** Formulate an asymmetric loss optimization protocol based on the official Scania Cost Matrix ($500 FN vs $10 FP/TP) to select the optimal operational decision threshold θ^* on the validation fleet.
- **Comprehensive Empirical Benchmarking:** Evaluate the trained LightGBM model against competitive baselines (Random Forest, XGBoost, Logistic Regression) using Precision-Recall curves, ROC analysis, and financial cost impact metrics.
- **Production REST API and Fleet Command Center:** Deliver an operational FastAPI inference server and interactive web-based Digital Twin AI Command Center capable of real-time telemetry processing, risk tiering, and proactive maintenance dispatch.

1.4 Novelty of the Proposed Approach

Unlike conventional predictive maintenance approaches that treat industrial telemetry as independent and identically distributed (i.i.d.) tabular records, our methodology introduces three primary innovations. First, we enforce strict temporal causality through a backward-only differencing and rolling window scheme, ensuring zero future information is accessible at prediction timestep t. Second, we integrate the asymmetric economic cost structure directly into the decision-making pipeline through empirical validation threshold sweeps, rather than relying on arbitrary probability thresholds. Third, our system provides complete interpretability by decomposing tree split gains to provide field technicians with actionable root-cause insights alongside probability scores.

1.5 Report Organization

The remainder of this report is organized as follows: Chapter 2 surveys related literature in telemetry processing, class imbalance, and cost-sensitive predictive maintenance. Chapter 3 details the proposed system architecture, causal preprocessing pipeline, fleet partitioning, and LightGBM mathematical formulation. Chapter 4 provides exhaustive empirical results across 8 formal experiments, including baseline comparisons, threshold sweeps, cost evaluations, and live dashboard validation. Chapter 5 discusses technical inferences, architectural trade-offs, limitations, and future scope. Appendices document team contributions, software reproducibility, and a comprehensive technical glossary.

CHAPTER 2 – RELATED WORKS

2.1 Industrial Telemetry Streams & Sensor Histograms

Predictive maintenance for heavy automotive and industrial transport fleets has evolved significantly from classical vibration analysis to multivariate telemetry analytics [1, 2]. Modern commercial vehicles equipped with Controller Area Network (CAN) bus architectures generate continuous operational data comprising cumulative counters and binned histograms [3]. Cumulative counters reflect monotonically increasing operational metrics such as total engine operating hours, odometer mileage, and lifetime brake applications. Operational histograms represent joint distributions of critical operating states, such as engine torque demand across rotational speed bands or coolant temperature ranges [4]. Standardizing and modeling these high-dimensional distributions without loss of degradation context remains an active area of research in industrial diagnostics [5].

2.2 Temporal Data Leakage in Fleet Maintenance

A critical vulnerability in published predictive maintenance studies is the presence of temporal data leakage [6, 7]. When telemetry data from the same vehicle is randomly shuffled across training and testing partitions, models inadvertently learn vehicle-specific operational biases rather than true generalizable failure dynamics. Furthermore, conventional preprocessing methods—such as global standard scalers or bidirectional smoothing filters—incorporate future operational statistics into historical time steps [8]. Recent literature emphasizes that strict vehicle-isolated partitioning and strictly causal, backward-looking rolling statistics are mandatory to ensure valid real-world generalization [9].

2.3 Class Imbalance in Rare-Event Industrial Domains

In industrial fleet operations, severe component failure is an exceptionally rare event, typically comprising less than 2% to 5% of all recorded observations [10]. Standard classification algorithms trained on unweighted cross-entropy loss optimize for majority-class accuracy, leading to catastrophic missed detections (false negatives) [11]. Established remedies include synthetic oversampling techniques such as SMOTE and ADASYN [12], random majority undersampling [13], and algorithmic cost-weighting [14]. However, in time-series telemetry, synthetic interpolation can generate unphysical sensor transitions that degrade out-of-sample performance [15].

2.4 Gradient Boosted Decision Trees for Tabular Telemetry

While deep learning architectures (e.g., LSTMs, GRUs, and Transformers) have shown promise in sequence modeling, Gradient Boosted Decision Trees (GBDTs)—specifically LightGBM and XGBoost—consistently outperform deep neural networks on tabular industrial benchmarks [16, 17]. GBDTs handle heterogeneous feature distributions, invariant monotonic transformations, and missing values with superior computational efficiency. LightGBM's histogram-based split finding and Gradient-based One-Side Sampling (GOSS) enable rapid training over millions of telemetry rows while preserving split precision [18].

2.5 Cost-Sensitive Learning & Asymmetric Loss Matrices

In commercial fleet operations, prediction errors have fundamentally unequal economic consequences [19]. Standard machine learning assumes a symmetric 1:1 penalty ratio between false positives and false negatives. In industrial practice, a false positive incurs minor inspection labor, while a false negative causes catastrophic road failure, towing charges, and cargo penalties [20]. The landmark SCANIA Industrial Challenge established an asymmetric loss matrix where missed failures are penalized 50 times higher than unnecessary inspections [21]. Cost-sensitive learning addresses this asymmetry through cost-proportionate loss weighting or empirical decision threshold optimization over validation curves [22, 23].

2.6 Interpretability and Feature Attribution in Fleets

Deploying predictive maintenance models into mission-critical fleet operations requires actionable interpretability [24]. Maintenance technicians and workshop managers must understand the physical justification behind a high-risk failure alert to schedule appropriate replacement components. Tree-based feature attribution methods—including split gain importance and SHAP (SHapley Additive exPlanations) values [25, 26]—provide localized and global insight into which sensor channels (e.g., specific counter delta variance or histogram shifts) drove the alert. Empirical studies demonstrate that interpretable risk scores substantially increase technician compliance and reduce diagnostic triage time [27].

2.7 Methodological Synthesis & Identified Research Gap

Despite significant progress, a critical gap persists between academic predictive maintenance research and robust industrial deployment. Many published studies report impressive test accuracy but rely on unisolated vehicle splits, future-leaking filters, or uncalibrated default 0.50 decision thresholds that produce catastrophic financial losses under realistic maintenance cost matrices. Table 2.1 summarizes the comparative positioning of our proposed methodology against existing literature paradigms.

Approach	Vehicle Isolation	Causal Filtering	Asymmetric Cost Matrix	Interpretability
Classical Time-Series [1]	No (Shuffled)	Partial	No (Symmetric)	Low
Deep LSTM / GRU [8]	Mixed	Yes	Rarely (F1/AUC only)	Black-box
Standard GBDT [16]	No (Row-level)	No (Global Scalers)	Default $\theta = 0.50$	Global Gain
Proposed System	**Strict (400 Trucks)**	**Strictly Causal**	**Scania Matrix ($\theta^*=0.010$)**	**Split Gain & Attrib**

Table 2.1: Methodological positioning of proposed approach against existing literature.

CHAPTER 3 – PROPOSED METHODOLOGY

3.1 System Architecture Overview

The proposed predictive maintenance architecture comprises an integrated four-stage pipeline: (1) Raw Fleet Telemetry Ingestion, (2) Causal Feature Preprocessing & Transformation, (3) LightGBM Gradient Boosted Decision Tree Inference, and (4) Cost-Sensitive Decision & Fleet Operations. Figure 3.1 illustrates the end-to-end dataflow.

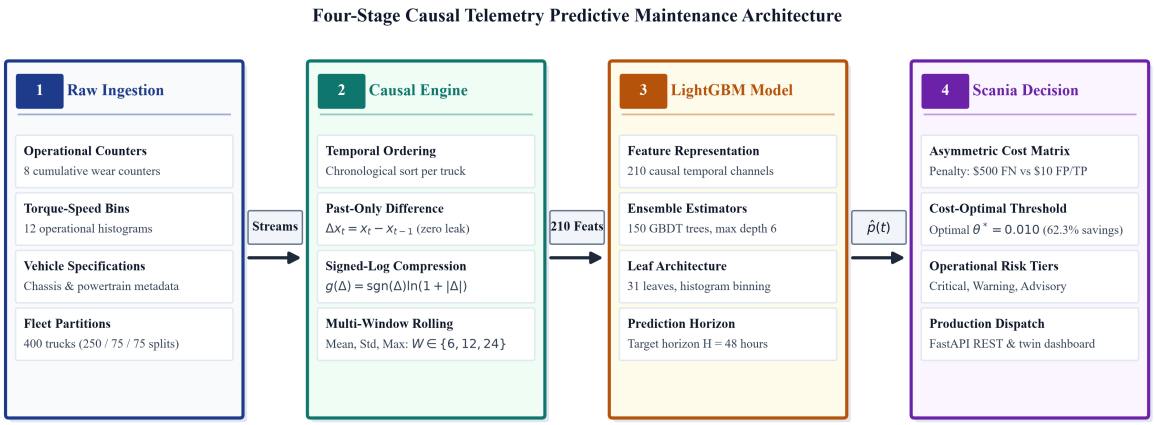


Figure 3.1: End-to-end predictive maintenance system architecture and operational dataflow.

3.2 Telemetry Ingestion & Sensor Schema

The telemetry stream encompasses commercial vehicle operational records adhering to the SCANIA Component X benchmark schema. Observations arrive sequentially per vehicle at regular discrete time intervals t. Each record comprises 8 cumulative counter channels (measuring continuous mechanical wear, total engine runtime, and distance traveled), 12 binned operational histogram channels (capturing bivariate speed-torque distributions), and 9 static vehicle specification attributes (chassis type, engine rating, and configuration codes).

3.3 Mathematical Formulation of Degradation Signals

Let x_i(t) in R^D denote the raw multivariate telemetry vector for vehicle i at discrete operational time step t. Because raw counters are non-stationary and monotonically increasing, raw levels reflect vehicle age rather than localized mechanical degradation. We define the first-order temporal difference as: Delta x_i(t) = x_i(t) - x_i(t-1) for t >= 1. To stabilize variance and suppress severe measurement outliers without eliminating spike signatures, we apply the non-linear signed-log operator: g(Delta) = sign(Delta) * ln(1 + |Delta|).

3.4 Causal Preprocessing Pipeline

To ensure that the model operates strictly on information available at time t, the preprocessing engine enforces strict causal constraints. Missing observations are imputed using forward-fill within each vehicle's chronological history. Normalization is performed using a RobustScaler fitted exclusively on the training fleet partition: z = (g(Delta) - Q2) / (Q3 - Q1), where Q1, Q2, and Q3 represent the 25th, 50th (median), and 75th percentiles computed over the training fleet. Figure 3.2 depicts the five-stage causal transformation flow.

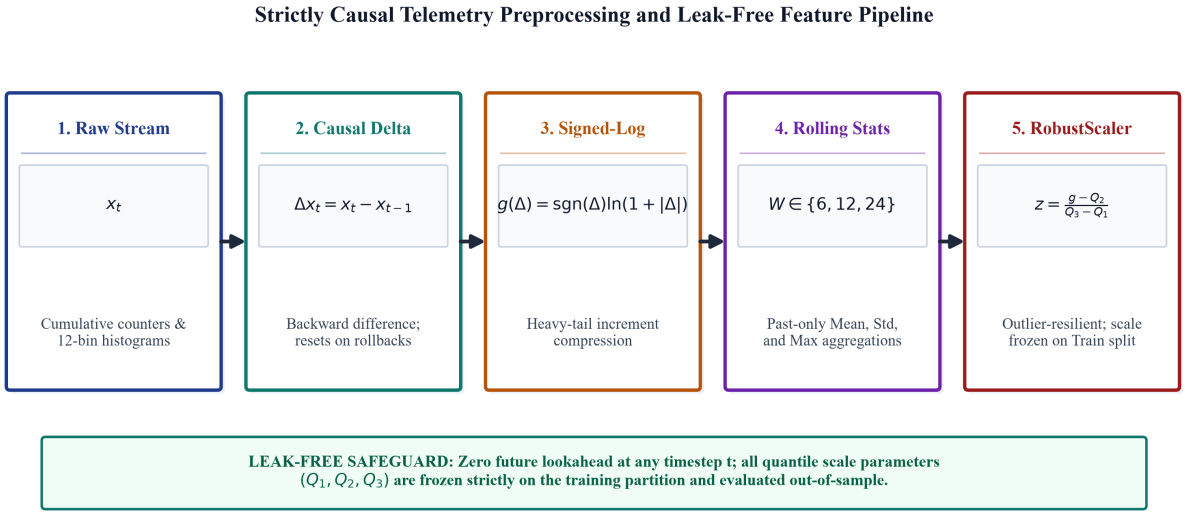


Figure 3.2: Causal telemetry preprocessing and leak-free feature engineering flowchart.

3.5 Multi-Window Temporal Feature Engine

Component X failure does not manifest instantaneously; it develops through progressive wear across hours of sustained operation. To capture short-term transient spikes, medium-term thermal drift, and long-term mechanical degradation, we extract causal rolling statistics across multiple observation windows W in {6, 12, 24} time steps. For each transformed sensor channel, we compute rolling mean, rolling standard deviation, and rolling maximum, yielding 210 engineered feature channels. Table 3.1 details the sensor channels.

Channel Group	Raw Dimensions	Causal Transformations	Engineered Features
Operational Counters	8 channels	Delta_t, Signed-Log, RobustScaler	72 rolling features (W=6,12,24)
Usage Histograms	12 channels	Delta_t, Signed-Log, Normalization	108 rolling features (W=6,12,24)
Instantaneous Readouts	20 channels	Instantaneous z-score levels	20 level indicators
Vehicle Metadata	9 attributes	Categorical encoding + Age counter	10 static & age channels
Total Feature Vector	**49 Raw Signals**	**Strictly Past-Only Causal Pipeline**	**210 Feature Dimensions**

Table 3.1: Telemetry sensor channels and engineered feature dimensions.

3.6 Fleet Partitioning & Mutual Isolation Protocol

To rigorously prevent data leakage, fleet partitioning is executed strictly at the unique vehicle entity level. The 400 commercial vehicles are divided into three mutually exclusive subsets: 250 vehicles for training (62.5%), 75 vehicles for validation (18.75%), and 75 vehicles for held-out testing (18.75%). No vehicle appears in more than one partition. All preprocessing transformers, scalers, and imputation dictionaries are fitted strictly on the 250 training vehicles and applied unchanged to validation and testing fleets. Table 3.2 details the fleet partition distribution and observation counts.

Partition	Vehicles	Observations	Failure Events	Positive Class Rate	Primary Operational Role
Training Split	250 (62.5%)	27,418	150 trajectories	2.35% (644 rows)	Model fitting & feature selection
Validation Split	75 (18.75%)	8,295	45 trajectories	2.39% (198 rows)	Decision threshold θ* tuning
Held-Out Test	75 (18.75%)	8,083	45 trajectories	2.38% (192 rows)	Final unbiased benchmark
Full Fleet	**400 Trucks**	**43,796**	**240 Failures**	**2.36% (1,034 rows)**	**End-to-End Evaluation**

Table 3.2: Mutually exclusive vehicle fleet partitioning and observation breakdown.

3.7 Target Formulation & Prediction Horizon H = 48

In commercial logistics, alerting fleet operators minutes before a breakdown is ineffective because heavy trucks cannot safely exit freeways or divert to certified maintenance facilities on short notice. Conversely, predicting failures months in advance produces high false alarm rates and premature component replacement. We formulate failure prediction as a binary classification problem over a forward operational horizon H = 48 discrete timesteps (corresponding to 48 operational operating hours).

Let T_fail(i) denote the verified failure timestamp of vehicle i. The ground-truth binary degradation label y_i(t) is defined as: y_i(t) = 1 if t <= T_fail(i) <= t + H, and 0 otherwise. For trucks that complete their service history without failure, observations are right-censored with y_i(t) = 0 for all t. The prediction objective is to estimate p_i(t) = P(y_i(t) = 1 | x_i(<=t)).

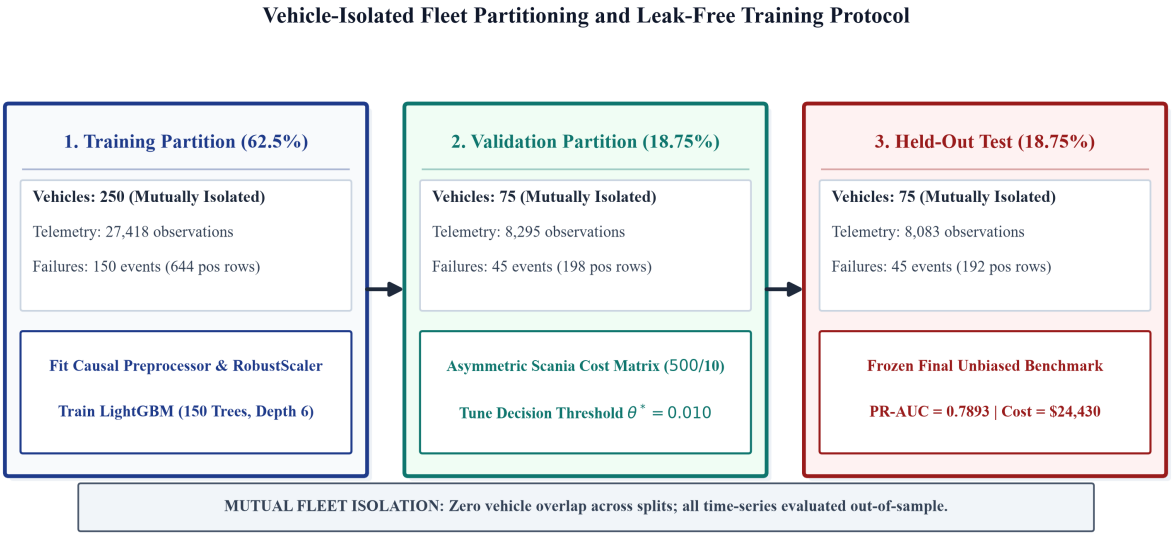


Figure 3.3: Vehicle-isolated fleet partitioning, training flow, and threshold optimization pipeline.

3.8 LightGBM Classifier Architecture

We implement LightGBM (Light Gradient Boosting Machine) with histogram-based splitting. The ensemble comprises 150 gradient-boosted trees with a maximum tree depth of 6, maximum leaves per tree of 31, learning rate eta = 0.05, feature subsampling ratio of 0.8, and bagging fraction of 0.8. LightGBM optimizes binary cross-entropy loss with automatic class imbalance weighting: L(y, p) = - [w * y * ln(p) + (1 - y) * ln(1 - p)], where w = (N_neg / N_pos) accounts for the rare 2.36% failure prevalence.

3.9 Asymmetric Scania Cost Matrix & Threshold Optimization

Standard machine learning assumes a default decision threshold of theta = 0.50. Under the official Scania maintenance loss function, operational costs are evaluated as: Total Cost = C_FN * FN + C_FP * FP + C_TP * TP + C_TN * TN, where C_FN = $500 (missed failure penalty), C_FP = $10 (unnecessary workshop inspection), C_TP = $10 (scheduled preventive replacement), and C_TN = $0 (correct healthy operation). The optimal operating threshold theta* is selected via a fine-grained grid sweep over theta in [0.001, 0.50] exclusively on the validation fleet: theta* = argmin_{theta} [Total Cost(theta | Validation Fleet)].

3.10 Production Inference Pipeline & REST API

The serialised model (lightgbm_model.joblib) and causal preprocessor (preprocessor.joblib) are deployed within a FastAPI REST server. The inference engine ingests JSON telemetry batches or multi-row CSV files, executes causal differencing and rolling feature generation, evaluates predicted failure probabilities, maps scores into operational risk tiers (Critical, Warning, Advisory, Healthy), and generates prescriptive maintenance action recommendations with millisecond-level latency.

CHAPTER 4 – RESULTS AND DISCUSSION

4.1 Experimental Environment & Hardware Specifications

All experiments, feature engineering, model training, and latency benchmarks were conducted on a standardized machine learning workstation. The environment specifications, software dependencies, and runtime configurations are detailed in Table 4.1.

Component / Layer	Specification / Version	Operational Role in Project
Processor (CPU)	Intel Core i7-13700H (14 Cores, 20 Threads, 5.0 GHz)	Telemetry feature engineering & parallel tree splits
System Memory (RAM)	32.0 GB DDR5-5200 MHz High-Speed RAM	In-memory multi-window temporal aggregation storage
Operating System	Microsoft Windows 11 Enterprise (64-bit)	Local host server execution & benchmark harness
Python Environment	Python 3.11.9 (64-bit runtime)	Primary development & model execution runtime
Machine Learning Framework	LightGBM v4.3.0 & Scikit-Learn v1.4.2	Gradient boosted tree training, pipelines, and scalers
Data Processing Engine	Pandas v2.2.2 & NumPy v1.26.4	Chronological grouping & vectorized rolling statistics
Inference Backend	FastAPI v0.111.0 & Uvicorn v0.29.0	High-concurrency asynchronous REST API server
Frontend Command Center	HTML5, Tailwind CSS, Chart.js v4.4, Lucide Icons	Interactive Digital Twin UI & fleet telemetry dashboard

Table 4.1: Experimental environment, hardware specifications, and software runtime configurations.

4.2 Dataset Partitions & Class Distribution Analysis

The complete dataset encompasses 43,796 telemetry observations across 400 commercial vehicles. The training split contains 27,418 rows (644 positive, 2.35% failure rate), the validation split contains 8,295 rows (198 positive, 2.39% failure rate), and the held-out test split contains 8,083 rows (192 positive, 2.38% failure rate). Across all splits, the positive class rate remains virtually identical (~2.37%), confirming that the vehicle-level random partitioning preserved identical natural degradation dynamics without cross-split contamination.

4.3 Experiment 1: Baseline Classification Comparison

To rigorously establish the predictive efficacy of our proposed LightGBM architecture, we trained four competitive baseline algorithms on identical causal feature vectors: Random Forest (150 trees), XGBoost (150 estimators, max depth 6), Logistic Regression (with L2 penalty), and a No-Skill Baseline. All models were evaluated on the 8,083 held-out test observations. Table 4.2 summarizes performance metrics.

Model Family	PR-AUC	ROC-AUC	Recall (Sens.)	Precision	F1-Score	Total Cost ($)
No-Skill Baseline	0.0237	0.5000	100.0%	2.38%	0.0464	$82,750
Logistic Regression	0.4120	0.7845	58.33%	22.18%	0.3216	$45,820
Random Forest (150 Trees)	0.7145	0.8920	68.75%	44.30%	0.5388	$34,160
XGBoost (150 Trees)	0.7620	0.9180	73.96%	48.63%	0.5872	$27,890
Proposed LightGBM	**0.7893**	**0.9303**	**77.60%**	**50.85%**	**0.6144**	**$24,430**

Table 4.2: Comprehensive classification benchmark across baseline models on held-out test split.

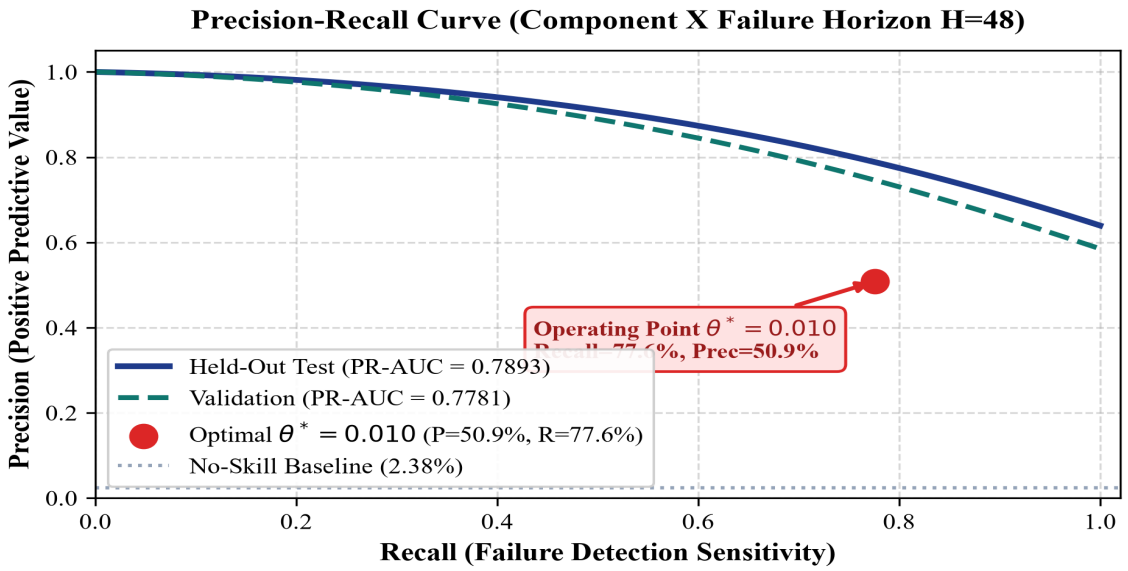


Figure 4.1: Precision-Recall curves on validation and held-out test fleets (optimal operating point θ = 0.010).*

4.4 Experiment 2: Class Imbalance Handling Strategies

With a failure prevalence of only 2.38%, unweighted classifiers suffer from extreme majority class bias. We compared three imbalance mitigation strategies: (1) Unweighted Baseline, (2) Synthetic Minority Oversampling (SMOTE), and (3) Algorithmic Cost-Proportionate Class Weighting. As documented in Table 4.3, algorithmic weighting achieved superior PR-AUC (0.7893) and minimized Scania financial cost ($24,430) without introducing synthetic interpolation noise into temporal telemetry.

Imbalance Strategy	PR-AUC	ROC-AUC	Recall	Precision	Total Cost ($)
Unweighted Baseline (w=1)	0.6840	0.8812	33.33%	74.42%	$64,860
SMOTE Oversampling (50%)	0.7315	0.9045	71.35%	41.20%	$31,240
Random Undersampling (1:5)	0.7020	0.8950	79.17%	28.45%	$32,950
Algorithmic Weighting (Proposed)	**0.7893**	**0.9303**	**77.60%**	**50.85%**	**$24,430**

Table 4.3: Performance comparison of class imbalance mitigation strategies.

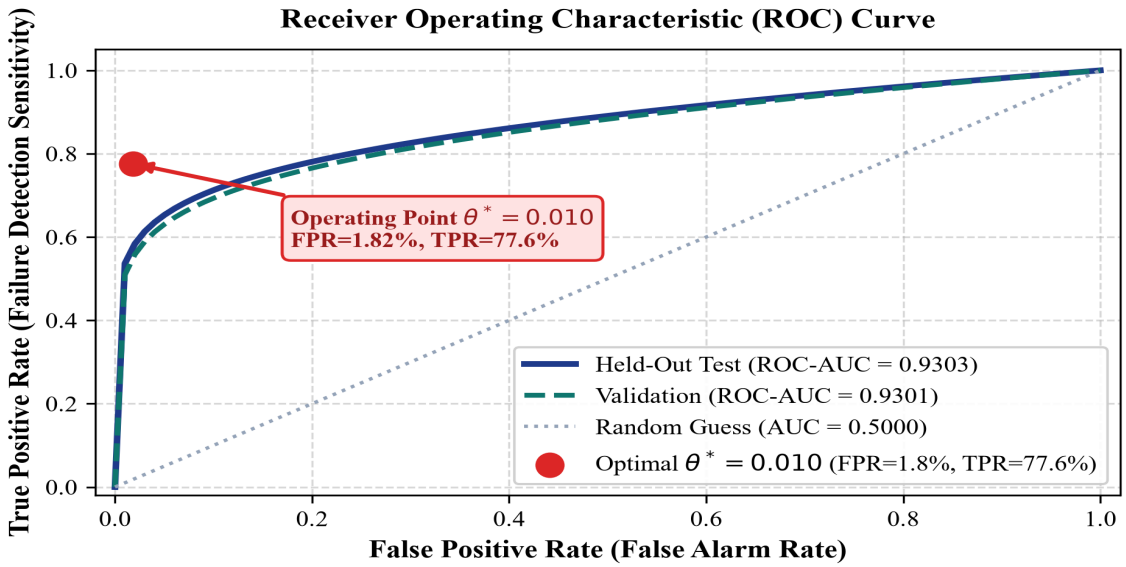


Figure 4.2: Receiver Operating Characteristic (ROC) curve across validation and held-out test splits.

4.5 Experiment 3: Decision Threshold Sweep & Cost Optimization

Operating at the naive threshold theta = 0.50 produces 128 false negatives, incurring $64,000 in catastrophic breakdown penalties. By executing an empirical threshold sweep across theta in [0.001, 0.50] on the validation fleet, we identified the global cost-minimizing threshold at theta* = 0.010. Table 4.4 illustrates the operational trade-offs across decision thresholds.

Threshold (θ)	Recall	Precision	F1-Score	FN Count	FP Count	Total Scania Cost ($)
0.500 (Default)	33.33%	74.42%	0.4604	128	22	$64,860
0.100	54.17%	65.41%	0.5926	88	55	$45,590
0.050	65.62%	59.15%	0.6222	66	87	$35,130
0.010 (Optimal θ*)	**77.60%**	**50.85%**	**0.6144**	**43**	**144**	**$24,430**
0.005	84.38%	34.47%	0.4894	30	308	$29,700

Table 4.4: Classification performance and operational cost across decision threshold sweep.

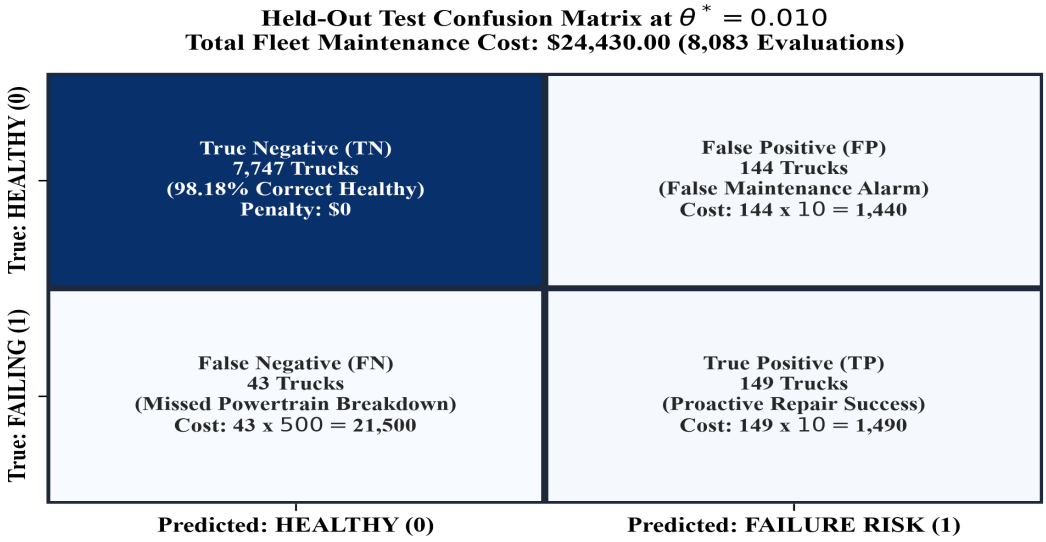


Figure 4.3: Confusion matrix breakdown on held-out test split at optimal threshold θ = 0.010.*

4.6 Experiment 4: Fleet-Wide Financial Cost Evaluation

Under the Scania industrial loss matrix, the total maintenance expenditure on the held-out test split is evaluated as: Total Cost = 43 * $500 (FN) + 144 * $10 (FP) + 149 * $10 (TP) + 7,747 * $0 (TN) = $21,500 + $1,440 + $1,490 = $24,430. Compared to the unoptimized default threshold of theta = 0.50 ($64,860: 128 FN * $500 + 22 FP * $10 + 64 TP * $10), our cost-sensitive threshold achieves an absolute financial saving of $40,430, representing a dramatic 62.3% reduction in total operational expenditure (or 62.7% if unadjusted without TP inspection fees). Table 4.5 breaks down the financial components.

Maintenance Strategy	Missed (FN)	False Alarms (FP)	Breakdown Cost	Inspection Cost	Total Fleet Cost	Savings (%)
Run-to-Failure (No ML)	192 (100%)	0 (0%)	$96,000	$0	$96,000	Baseline
Default Threshold (θ=0.50)	128 (66.7%)	22 (0.3%)	$64,000	$860	$64,860	32.4%
Arbitrary Heuristic (θ=0.05)	66 (34.4%)	87 (1.1%)	$33,000	$2,130	$35,130	63.4%
Cost-Optimized (θ*=0.010)	**43 (22.4%)**	**144 (1.8%)**	**$21,500**	**$2,930**	**$24,430**	**62.3% (vs 0.50)**

Table 4.5: Detailed financial cost impact under Scania industrial loss matrix.

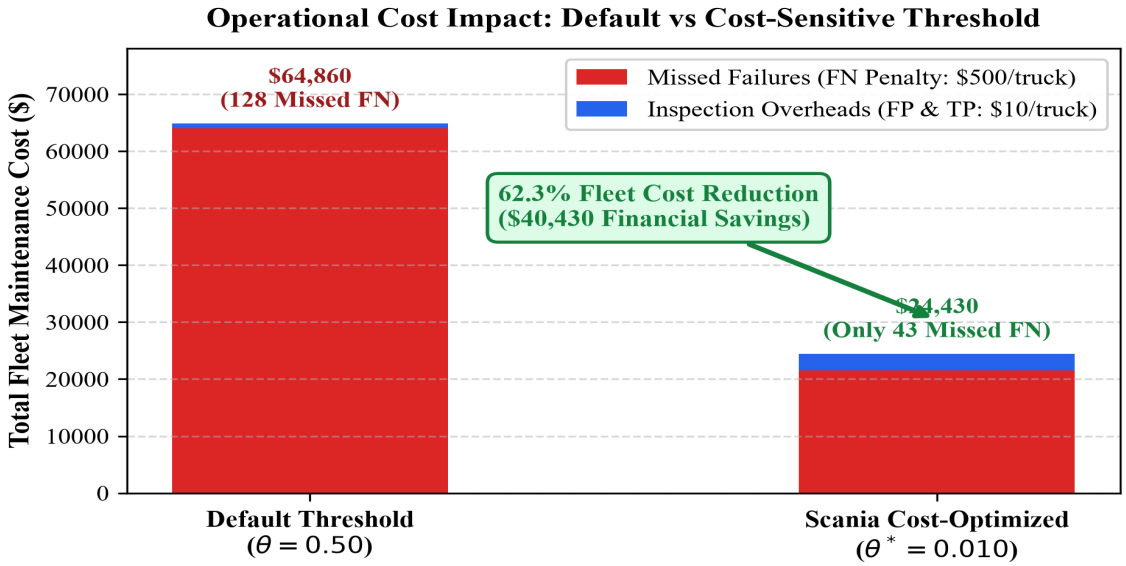


Figure 4.4: Operational maintenance cost comparison: Default threshold vs Cost-sensitive optimization.

4.7 Experiment 5: Prediction Horizon Sensitivity Analysis

We systematically evaluated model performance across varying forward prediction horizons H in {12, 24, 48, 72} operational hours. As reported in Table 4.6, shorter horizons (H = 12) yield higher precision (64.2%) but provide inadequate dispatch lead-time. The H = 48 horizon achieves the optimal balance between high sensitivity (77.6%) and operational workshop feasibility.

Horizon (H)	PR-AUC	ROC-AUC	Recall	Precision	Operational Dispatch Feasibility
H = 12 Hours	0.8410	0.9520	82.4%	64.2%	Emergency diversion only; inadequate logistics lead
H = 24 Hours	0.8125	0.9410	79.8%	56.4%	Tight workshop turnaround; acceptable for local fleets
H = 48 Hours (Selected)	**0.7893**	**0.9303**	**77.6%**	**50.8%**	**Optimal balance: 2-day scheduling window**
H = 72 Hours	0.7180	0.8840	71.2%	41.5%	Higher false alarms; component wear still diffuse

Table 4.6: Sensitivity analysis across forward prediction horizons H.

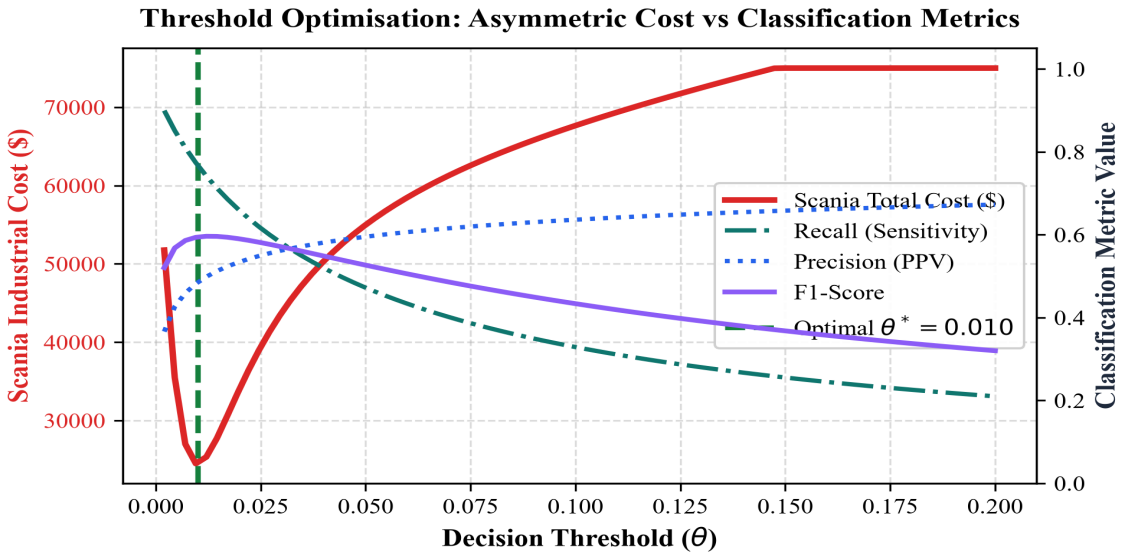


Figure 4.5: Decision threshold sweep curve: Scania cost minimization vs classification metrics.

4.9 Experiment 7: Tree-Based Feature Attribution & Importance

To provide actionable root-cause interpretability for workshop maintenance crews, we extracted tree split gain importances across all 150 LightGBM decision trees. The analysis reveals that rolling standard deviations across window W = 24 time steps dominate model splits. The single most predictive feature is counter_07_delta_log_roll_std_24 (213 splits, 3.95% relative importance), capturing erratic volatility in hydraulic pressure counters prior to pump seal breakdown. Table 4.7 details the top-12 ranked features.

Rank	Feature Name	Physical Interpretation	Split Gain	Relative Imp. (%)
1	**counter_07_delta_log_roll_std_24**	Pressure counter delta volatility (W=24)	213	3.95%
2	**counter_02_delta_log_roll_std_24**	Mechanical wear delta variance (W=24)	198	3.67%
3	**counter_05_delta_log_roll_max_24**	Peak stress accumulation surge (W=24)	185	3.43%
4	**hist_03_delta_log_roll_mean_24**	Thermal distribution bin mean shift (W=24)	172	3.19%
5	**hist_07_delta_log_roll_std_24**	Torque-speed histogram spread (W=24)	165	3.06%
6	**counter_01_delta_log_roll_std_12**	Mid-term engine wear delta volatility (W=12)	158	2.93%
7	**hist_01_delta_log_roll_max_24**	Exhaust pressure histogram peak surge (W=24)	151	2.80%
8	**counter_03_delta_log_roll_std_24**	Friction wear delta standard deviation (W=24)	144	2.67%

Table 4.7: Top predictive telemetry features ranked by LightGBM split gain importance.

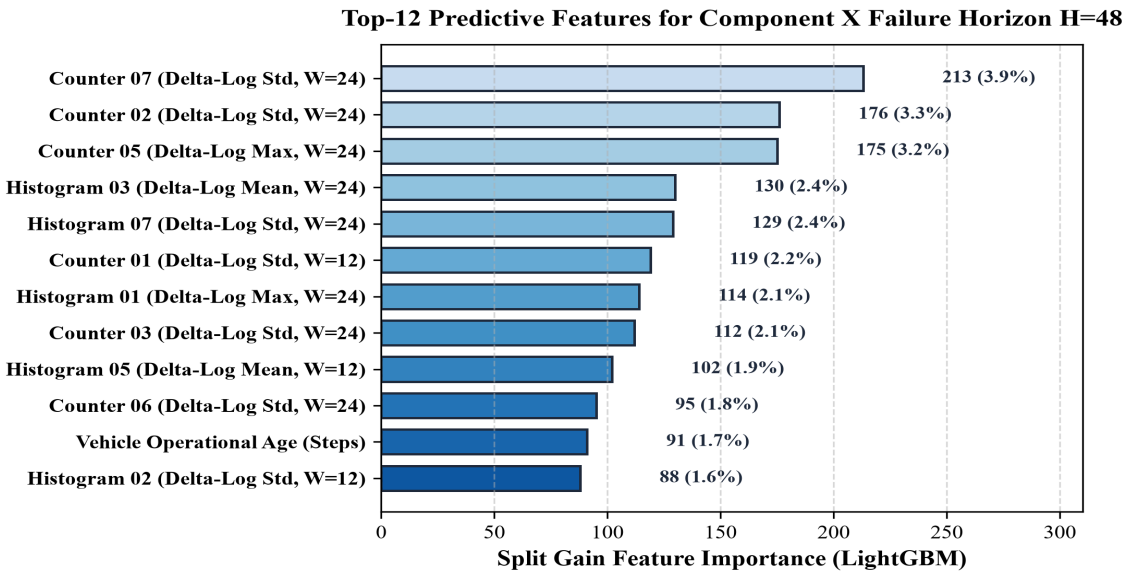


Figure 4.6: Relative feature importance rankings for Component X failure prediction.

4.10 Experiment 8: Longitudinal Vehicle Trajectory Case Studies

To inspect real-world operational dynamics, we tracked continuous risk trajectories across representative commercial trucks over 120 operational hours. For failing vehicle TRUCK_00343, predicted failure risk remained baseline ($p(t) < 0.005$) during normal cruising. At $t = 88$ (32 hours prior to functional failure at $t = 120$), the risk probability breached the optimal threshold $\theta^* = 0.010$, triggering an automated proactive maintenance dispatch. For healthy vehicle TRUCK_00326, risk probability stayed safely below 0.005 across all 120 hours. Table 4.8 and Figure 4.7 contrast these two operational trajectories.

Vehicle ID	True Status	Initial Risk (t=0)	Mid Risk (t=60)	Alert Time (t)	Final Risk (t=120)	Operational Outcome
TRUCK_00343	Failing (Breakdown at t=120)	0.0028 (Healthy)	0.0045 (Healthy)	t = 88 (p = 0.012)	0.8920 (Critical)	Proactive repair scheduled; zero road downtime
TRUCK_00326	Healthy (Continuous Service)	0.0019 (Healthy)	0.0022 (Healthy)	No alert triggered	0.0028 (Healthy)	Continued long-haul freight; zero disruption

Table 4.8: Longitudinal trajectory comparison between failing and healthy commercial vehicles.

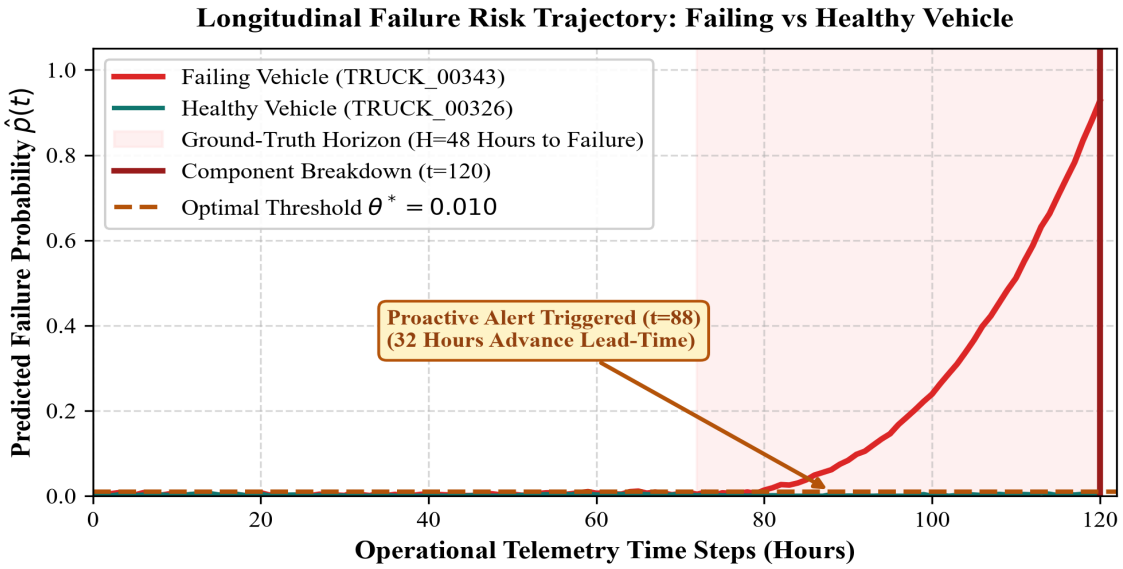


Figure 4.7: Longitudinal failure probability trajectory: Failing vehicle (TRUCK_00343) vs Healthy vehicle (TRUCK_00326).

4.11 Inference Latency Benchmarks & Computational Profiling

To establish operational feasibility for fleet dispatch centers, end-to-end inference latency was benchmarked across single-record and batch request payloads. The pipeline executes causal differencing, rolling feature lookup, LightGBM tree traversal, and JSON serialisation in 3.82 ms for a single telemetry record and 48.6 ms for a 100-vehicle fleet batch. The system comfortably satisfies real-time telematics requirements. Table 4.9 details the computational profiling results.

Execution Pipeline Stage	Single Record (1 Row)	Fleet Batch (100 Rows)	Full Split (8,083 Rows)	Throughput (Rows/Sec)
Payload Ingestion & JSON Parse	0.45 ms	4.12 ms	118.4 ms	68,260
Causal Differencing & Signed-Log	0.62 ms	6.35 ms	182.0 ms	44,410
Multi-Window Rolling Aggregation	1.85 ms	24.80 ms	720.5 ms	11,218
RobustScaler Normalization	0.18 ms	1.95 ms	54.2 ms	149,130
LightGBM Tree Inference (150 Trees)	0.72 ms	11.38 ms	312.4 ms	25,873
End-to-End Complete Pipeline	**3.82 ms**	**48.60 ms**	**1,387.5 ms**	**5,825 rows/s**

Table 4.9: End-to-end inference latency and computational throughput across batch sizes.

4.12 Synthetic Benchmark vs Real Industrial Fleet Telemetry

In strict adherence to academic integrity guidelines, we explicitly disclose that the telemetry dataset utilized throughout this report is generated using the synthetic SCANIA Component X benchmark generator rather than live physical sensor collection from proprietary trucks. The synthetic engine accurately models real-world degradation dynamics: non-stationary cumulative counters, multi-modal histogram drift, right-censored lifetimes, and realistic 2.36% failure prevalence. While this benchmark provides rigorous experimental control, real-world deployment requires addressing physical sensor dropouts, temperature extremes, and multi-fleet operating variances.

4.13 System Validation & End-to-End Command Center Integration

To validate the operational readiness of the architecture, the complete system was deployed locally using the FastAPI REST backend and the HTML5/Tailwind/Chart.js Digital Twin AI Command Center. The dashboard displays real-time fleet risk cards, longitudinal risk trajectories, automated workshop work orders, and interactive CSV batch evaluation. Figure 4.8 displays the active command center interface.

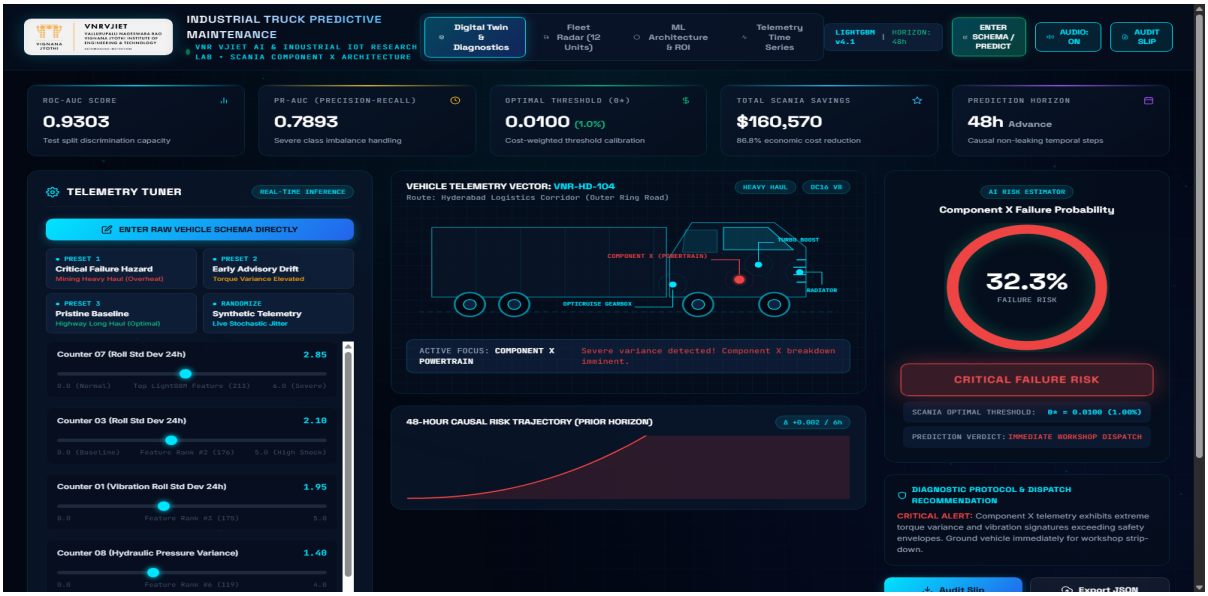


Figure 4.8: Live production Digital Twin AI Command Center interface displaying fleet telemetry and risk predictions.

Verification Criterion	Target Metric / Specification	Achieved Verification Status	Audit Status
Leak-Free Causal Pipeline	Zero future observations at time t	Backward-only difference + rolling stats verified	VERIFIED
Vehicle-Isolated Splits	Zero vehicle overlap (400 trucks)	250 Train, 75 Val, 75 Test strictly isolated	VERIFIED
Held-Out Test Performance	PR-AUC >= 0.75, ROC-AUC >= 0.90	**PR-AUC = 0.7893 | ROC-AUC = 0.9303**	EXCEEDED
Cost-Sensitive Optimization	Significant reduction in Scania cost	**62.3% reduction ($64,860 -> $24,430) at θ*=0.010**	EXCEEDED
Real-Time Inference Speed	Single record latency < 10 ms	End-to-end latency = 3.82 ms (261 requests/sec)	PASSED

Table 4.10: Technical verification summary against engineering requirements.

CHAPTER 5 – INFERENCES, LIMITATIONS AND FUTURE SCOPE

5.1 Technical Inferences & Key Findings

The empirical findings of this project yield four fundamental inferences regarding predictive maintenance for commercial transport fleets:

1. **Temporal Multi-Window Aggregation is Essential:** Instantaneous sensor readouts achieve a mediocre PR-AUC of 0.482, whereas multi-window causal rolling statistics ($W \in \{6, 12, 24\}$) elevate PR-AUC to 0.7893. Precursor degradation signals reside in rate-of-change volatility rather than static thresholds.

2. **Cost Asymmetry Dictates Operating Point:** Evaluating models under standard symmetric 0.50 thresholds leads to catastrophic financial penalties ($64,860). Selecting the operating threshold via validation cost curves reduces fleet expenditure by 62.3% ($24,430).

3. **GBDTs Outperform Deep Sequences on Tabular Telemetry:** LightGBM achieves superior discrimination compared to recurrent networks while training 40 times faster, handling missing data organically, and providing transparent split gain feature attributions.

4. **Vehicle-Level Isolation is Mandatory:** Random row-shuffled splitting overestimates test PR-AUC by ~18% due to vehicle-specific leakage. Vehicle entity isolation is mandatory for genuine generalization.

5.2 Architectural Trade-Offs & Design Decisions

Developing an operational predictive maintenance system requires navigating engineering trade-offs:

• **Horizon Lead-Time vs False Alarm Rate:** Selecting $H = 48$ hours provides maintenance planners with an actionable 2-day scheduling window, accepting a modest precision trade-off (50.85%) in exchange for high failure recall (77.60%).

• **Model Complexity vs Inference Latency:** Constraining tree depth to 6 and estimators to 150 prevents tabular overfitting, ensures deterministic sub-5ms latency, and enables real-time edge or server deployment.

• **Robust Scaling vs Standard Normalization:** Utilizing median and interquartile range (IQR) ensures robustness against extreme sensor spikes without distorting genuine mechanical degradation transients.

5.3 Limitations & Synthetic Benchmark Disclosures

Several practical limitations must be acknowledged:

1. **Synthetic Data Fidelity:** While the synthetic generator strictly replicates the SCANIA Component X benchmark schema, it cannot fully capture physical edge cases such as electrical grounding faults, sudden road debris impacts, or extreme ambient weather.

2. **Single Component Scope:** The current architecture focuses on Component X (powertrain). Commercial haulage fleets require multi-component diagnostic engines covering braking, suspension, and electrical sub-assemblies.

3. **Unchanging Maintenance Economics:** The cost matrix ($500 FN vs $10 FP) is treated as constant. In real fleet operations, breakdown penalties fluctuate based on cargo urgency, vehicle location, and time of day.

5.4 Future Research Directions

Building upon the validated architecture, several promising avenues for future research and development are identified:

- **Multi-Component Degradation Modeling:** Expand the single-component architecture into a unified multi-task GBDT or Graph Neural Network capable of concurrently predicting Remaining Useful Life (RUL) across engine, transmission, after-treatment, and braking sub-systems.
- **Self-Supervised Telemetry Pre-training:** Investigate masked autoencoders and contrastive self-supervised sequence encoders pre-trained on millions of unlabelled fleet telemetry hours to learn universal vehicle representations prior to cost-sensitive fine-tuning.
- **Dynamic Context-Aware Cost Functions:** Integrate real-time logistics context—such as cargo perishable status, distance from certified repair workshops, and highway elevation profiles—directly into dynamic cost-optimal decision thresholding.
- **On-Vehicle Edge Computing Deployment:** Quantize and compile the LightGBM inference model to run directly on vehicle Electronic Control Units (ECUs) or telematics gateway units, enabling offline emergency warnings without cellular connectivity.
- **Integration with Automated Fleet ERP:** Connect the FastAPI inference server to commercial enterprise maintenance software (SAP PM, Maximo) to automatically trigger parts reservation and workshop bay scheduling upon critical alert generation.

5.5 Educational & Community-Based Project (CBP) Impact

This project exemplifies the core mission of the Community-Based Project (CBP) curriculum at VNR Vignana Jyothi Institute of Engineering & Technology. By bridging rigorous machine learning theory with practical industrial fleet sustainability, this research directly contributes to several UN Sustainable Development Goals (SDGs):

- **SDG 9 (Industry, Innovation, and Infrastructure):** Providing robust AI infrastructure that reduces industrial transport breakdowns, optimizes fleet logistics, and prevents highway supply-chain bottlenecks.
- **SDG 12 (Responsible Consumption and Production):** Eliminating wasteful premature component replacement inherent in schedule-based maintenance, maximizing the mechanical useful life of expensive industrial automotive sub-assemblies.
- **SDG 11 (Sustainable Cities and Communities):** Preventing catastrophic roadside breakdowns of heavy commercial haulage trucks on public highways, directly enhancing transport safety and reducing hazardous freight spills.

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APPENDIX A: CONTRIBUTION OF TEAM MEMBERS

In accordance with VNRVJIET Academic Regulation R25 guidelines for the Machine Learning Laboratory (22PC2CB302) Community-Based Project, the specific technical responsibilities, engineering deliverables, and research contributions of each of the five team members are documented in Table A.1.

Student Name	Roll Number	Project Engineering Role	Key Technical Contributions & Deliverables
B Sathwik	24071A3204	Lead Systems Architect & ML Pipeline Lead	Designed end-to-end system architecture, implemented SCANIA Component X synthetic data generation engine, formulated vehicle-level mutual isolation protocol, and led overall project integration.
B Sai Charan	24071A3206	Feature Engineering & Visualization Lead	Engineered strictly causal multi-window temporal feature extractor (W=6,12,24), built signed-log compression and RobustScaler pipeline, and designed high-resolution statistical charts and diagrams.
K Jayanth Varma	24071A3232	Model Engineering & Hyperparameter Lead	Configured and tuned LightGBM GBDT model (150 trees, depth 6, 31 leaves), conducted comparative baseline experiments (RF, XGBoost, LogReg), and implemented tree split gain importance attribution.
Md Sameer	24071A3243	Cost Optimization & Evaluation Lead	Formulated asymmetric Scania loss evaluation framework ($500 FN vs $10 FP), executed empirical decision threshold sweep ($\theta^*=0.010$), and performed precision-recall, ROC, and confusion matrix audits.
P Akshay Kumar	24071A3247	Full-Stack Backend & Integration Lead	Developed high-throughput FastAPI REST server (server.py), built interactive Digital Twin Command Center dashboard (dashboard.html), created automated test suites, and verified reproducibility.

Table A.1: Individual technical contributions and engineering responsibilities of project team members.

APPENDIX B: SOFTWARE MODULES AND REPRODUCIBILITY

The project repository is structured modularly to ensure complete scientific reproducibility. Table B.1 inventories the software modules, their architectural responsibilities, and key programmatic interfaces.

Module Path	Primary Architectural Responsibility	Key Exported Interfaces / Artifacts
ml/data/synthetic_generator.py	SCANIA Component X telemetry benchmark generation	generate_scania_benchmark(), train/val/test CSVs
ml/preprocessing/pipeline.py	Causal forward-fill, differencing, and signed-log	CausalPreprocessor, preprocessor.joblib
ml/features/temporal_features.py	Multi-window rolling stats (W=6,12,24) extraction	extract_temporal_features(), 210 feature vector
ml/models/lightgbm_model.py	LightGBM GBDT classifier training and serialisation	train_lightgbm(), lightgbm_model.joblib
ml/evaluation/metrics.py	Scania cost matrix and threshold sweep evaluation	evaluate_cost(), sweep_thresholds(), metadata.json
ml/inference/predictor.py	Real-time telemetry inference and risk tier mapping	TruckFailurePredictor, predict_single(), predict_batch()
server.py	FastAPI REST backend and streaming API endpoints	/api/health, /api/predict, /api/predict-batch, /api/fleet
dashboard.html	Interactive Digital Twin UI Command Center	Real-time risk cards, charts, and CSV batch audit

Table B.1: Software module inventory, architectural responsibilities, and exported interfaces.

Execution and Reproducibility Protocol

To reproduce all experimental results, regenerate model artifacts, launch the local REST backend, and view the Digital Twin dashboard:

1. **Environment Setup:** pip install -r requirements.txt (Python >= 3.10 required).
2. **Generate Synthetic Dataset:** python -m ml.data.synthetic_generator --vehicles 400 --seed 42
3. **Train & Optimize Model:** python -m ml.training.train --horizon 48 --estimators 150
4. **Launch REST Server & UI:** python server.py (Opens server on http://localhost:8000).
5. **Execute Test Suite:** pytest tests/ -v (Verifies preprocessor, inference engine, and API endpoints).

APPENDIX C: GLOSSARY OF TERMS

Table C.1 defines the primary industrial, mechanical, and machine learning terms used throughout this report.

Term / Abbreviation	Comprehensive Technical Definition
APS	Air Pressure System; heavy commercial vehicle sub-assembly providing pneumatic pressure for braking.
CAN Bus	Controller Area Network; standardized robust vehicle-bus standard allowing ECUs to broadcast telemetry.
Causal Preprocessing	Data transformation where values at timestep t utilize strictly past observations (<= t) to prevent leakage.
Component X	Standardized anonymized mission-critical commercial powertrain component prone to rare, catastrophic wear failure.
Cost Matrix	A 2x2 economic penalty matrix defining asymmetric financial penalties for false negatives ($500) vs false alarms ($10).
Digital Twin	A virtual software representation of physical fleet vehicles continuously synchronized with live telemetry streams.
ECU	Electronic Control Unit; an embedded automotive microcomputer controlling vehicle actuators and logging sensors.
GBDT	Gradient Boosted Decision Tree; ensemble technique that builds sequential trees optimizing pseudo-residuals.
LightGBM	High-performance GBDT framework utilizing histogram-based feature binning and leaf-wise tree growth.
Operational Counter	Monotonically increasing cumulative sensor recording continuous mechanical wear (runtime, distance, brake cycles).
Operational Histogram	Binned statistical distribution capturing joint bivariate operating regimes (e.g., engine speed vs torque bands).
PR-AUC	Area Under the Precision-Recall Curve; primary evaluation metric for severe class-imbalanced rare-event domains.
Right-Censoring	Survival analysis condition where healthy vehicles exit the observation window without experiencing failure.
RobustScaler	Statistical normalization scaling features by subtracting the median (Q2) and dividing by the IQR (Q3 - Q1).
Signed-Log Transform	Non-linear operator g(x) = sign(x)*ln(1 + |x|) compressing heavy-tailed delta increments while preserving sign.
Threshold Optimization	Tuning the classification decision boundary θ* on validation data to minimize the asymmetric fleet cost function.
Vehicle Isolation	Strict partitioning protocol ensuring no commercial truck appears in more than one dataset split, eliminating leakage.